MIL-T-12712 (ORD)
13 May 1953
SUPERSEDING
Army 51-72-68
13 December 1948

MILITARY SPECIFICATION

TELESCOPE, M35,(SERIES)

This specification consists of this cover sheet and Specification 51-72-68, dated 13 December 1948, together with Amendment 1, dated 14 April 1949, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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U. S. ARMY
SPECIFICATION

No. 51-72-68

13 DECEMBER 1948

TELESCOPE, M85 (SERIES)

A. APPLICABLE SPECIFICATIONS

A–1. Ordnance Department Drawing Class 91, Division 260 Drawing 1, titled, "Telescope M85, List of Drawings", lists or indicates all drawings and specifications which form part of this specification.

A–2. All drawings and specifications shall be of the issue in effect on date of invitation for bids. The issue in effect shall be that issue which the contracting officer shall designate or furnish the bidder.

B. TYPES

B–1. This specification covers a series of telescopes known basically as Telescope, M85. The individual telescope has a designation affixed to the model number to identify the type reticle employed.

C. MATERIAL AND WORKMANSHIP

C–1. *Material shall* be in accordance with the drawings, general specifications, and material specifications forming part of this specification.

C–2. *Workmanship* shall be in accordance with U. S. Army Specification No. 51–7, Instruments, Fire Control; General Specification Governing the Manufacture and Inspection of, and U. S. Army Specification No. 51–70–2, Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of.

D. GENERAL REQUIREMENTS

D–1. *General requirements* and associated tests common to all optical fire control instruments are contained in U. S. Army Specifications No. 51–7 and 51–70–2 and are mandatory in so far as they are applicable.

E. DETAIL REQUIREMENTS

E–1. *Definition.*—The image shall be practically free from distortion, astigmatism, curvature of field, coma and chromatic and spherical aberrations, consistent with the design for a radial distance of 2° 45′ from the center of the field of view. Images showing any of these defects in the area specified shall be cause for rejection of the telescope, if judged objectionable by the inspector.

E–1a. A telescope chosen from the initial production on the contract, representing the lower limit of acceptability of image quality, as determined by the contracting officer, shall be standard for inspection purposes.

E–2. *Parallax.*—When viewing a target within the range of distances from 125 to 150 yards there shall be no parallax between the image and the reticle at the center of the field.

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E–3. *Collimation.*—The optical axis shall not be out of coincidence with the geometric axis of the telescope, as established by the exterior mounting surfaces, by more than 0.25 mil.

E–4. *Open sights.*—The line of sight of the open sight shall be parallel with the optical axis of the telescope within 1°.

E–5. *Plumb of reticle.*—With the telescope mounting surfaces in a horizontal plane, the 800-yard horizontal reticle line shall not deviate from the image of a horizontal target reference line by more than the angle subtended by 0.2 mil block at the extremity of the horizontal reticle line when the other extremity of this line is in coincidence with the reference line.

E–6. *Image tilt.*—The image of a plumb line shall be within 1° of verticality as measured by an auxiliary telescope in front of the eyepiece.

E–7. *Reticle accuracy.*—The angular values between the zero range cross and the various range marks shall be as indicated on the reticle drawing within 2 percent of the elevation in mils plus 0.25 mil.

E–8. *Illumination.*—With the reticle illuminated by Instrument Light, M36 or equivalent, the reticle markings shall appear clearly defined when observed in a dark room.

E–9. *Waterproofing.*—The telescope shall be properly sealed against the entrance of moisture into its interior.

E–10. *Temperature* requirements specified in U. S. Army Specification No. 51–7 for Type II instruments shall apply.

E–11. *Vibration.*—Each telescope shall be subjected to the vibration requirements of U. S. Army Specification No. 51–70–2.

F. METHODS OF INSPECTION AND TESTS

F–1. Unless otherwise specified each telescope and each spare part and assembly shall be inspected for conformance with this specification and the applicable requirements of U. S. Army Specifications No. 51–7 and No. 51–70–2.

F–2. *Waterproofing tests.*—The arrangement for the testing apparatus is shown in figure below:

V_1 = Airtight container whose volume is equal to the displacement volume of the telescope.

V_2 = Airtight container whose volume (when empty) is equal to twice the displacement volume of the telescope.

(1) Place telescope in airtight container V_2.
(2) Close valve C_2.
(3) Open valve C_1 until M_1 indicates 10 pounds per square inch plus or minus 1. Close valve C_1.
(4) Open valve C_2. The pressure in the combined volumes V_1 and V_2 (with telescope) shall be 5 pounds per square inch plus or minus 0.5 as indicated by M_2.

(5) As soon as oscillations of M_1 cease, close valve C_2.
(6) As soon as possible after closing valve C_2 note pressure as indicated by M_2. One minute after noting pressure of M_2, again note the pressure of M_2.
(7) There shall be no difference between the two pressure readings.

F–3. Temperature test specified in U. S. Army Specification No. 51–7 for Type II instruments shall apply.

G. PACKING AND MARKING

G–1. *Packing.*—

G–1a. Packaging, packing, and marking shall be as specified in U. S. Army Specification No. 51–7.

H. NOTES

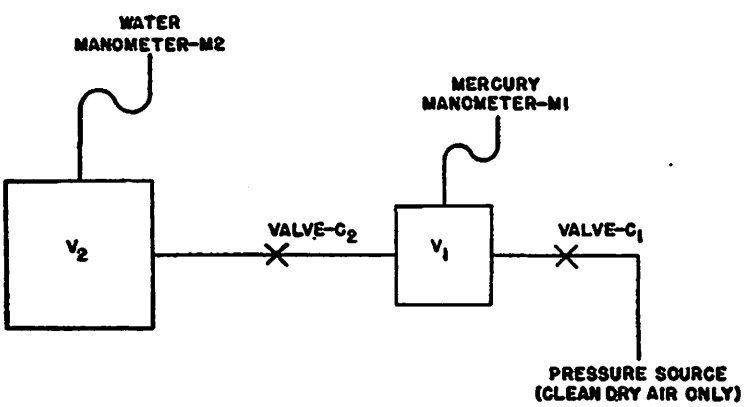
H–1. This specification replaces Frankford Arsenal Tentative Specification FXS–1008.

H–2. The use of this specification whenever applicable is mandatory on all procuring agencies of the Army.

Notice.—This specification, together with specifications and drawings pertaining to it and bearing a "NOTICE" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Note.—Copies of this specification may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

[No. 51–72–68]



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